

TSA Pipeline Security Directive — Compliance Report

Cisco Secure Workload | IT-Side Mapping for Natural Gas and Oil
Pipeline Operators

Cisco Secure Workload — Compliance Mapping Series

Version 1.0 (draft — pending SME review)

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Cisco Secure Workload — TSA Pipeline Security Compliance Mapping

How to read this report

This report describes how Cisco Secure Workload (CSW) supports a TSA-designated pipeline owner/operator's compliance posture against the TSA Security Directive Pipeline-2021-02 series cybersecurity requirements on the **IT side** of the IT/OT boundary.

Scope statement up front. CSW is a workload-protection and segmentation platform for IT estate — servers, virtual machines, containers, and cloud workloads. CSW does **not** enforce policy on RTUs, flow computers, PLCs, valve actuators, or any other Level 0–2 OT device on the pipeline itself. CSW's TSA SD contribution is at the **IT-side of the IT/OT boundary** and within the IT systems that support pipeline operations — engineering workstations, SCADA jump servers, historian databases, MES/EAM systems, vendor remote- access hosts, identity/PKI services, and the corporate IT systems that touch Critical Cyber Systems on the IT side.

Companion document. The accompanying *TSA Pipeline Reference Design* walks through deployment, scoping, IT/OT segmentation design, access control, continuous monitoring, vulnerability management on the IT side, the Cybersecurity Incident Response Plan evidence pack, and the Cybersecurity Assessment Plan inputs. Read this report for the audit-narrative view; read the reference design for the deployment view.

Note on revision letters. TSA has revised SD Pipeline-2021-02 multiple times (02, 02A, 02B, 02C, 02D, ...). The substantive controls in the III.A–III.D sections have been stable in shape but have been refined in detail. This report is written generically to the current effective revision; reconcile against the active version on the TSA web site before relying on this document in an inspection response.

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

1. Executive Summary

The TSA Pipeline Security Directive 2021-02 series imposes mandatory cybersecurity requirements on TSA-designated Critical Pipeline owner/operators in the United States. The directive's substantive cybersecurity content sits in four sections (III.A through III.D) that line up cleanly with the IT-side capabilities of a workload- protection platform: segmentation, access control, continuous monitoring, and reduction of unpatched-system risk.

What CSW changes for a TSA SD programme:

- **Section III.A — IT/OT segmentation evidence.** The IT/OT firewall remains your boundary, but CSW enforces deny-by-default *up to* that boundary on the IT side. Undocumented IT pivots toward the IT/OT boundary are structurally removed.
- **Section III.A — Documented data flows.** CSW provides the observed-vs-documented IT → OT flow inventory that lets architecture reviews focus on gaps and changes rather than rebuilding the as-is picture.
- **Section III.B — Access control evidence.** Workload-level least- privilege with quarterly access review and sunset logs for unused rules.
- **Section III.C — Continuous monitoring.** Behavioural detections, conversation-graph anomalies, and per-flow telemetry to your SIEM, paired with your OT-aware sensor stack for end-to-end coverage.
- **Section III.D — Unpatched-system risk reduction.** Continuous CVE inventory per workload with CVSS, exploit availability, and EPSS context; compensating controls registered for delayed patches.
- **CIRP and CAP evidence.** Six-artefact incident reconstruction bundles for your Cybersecurity Incident Response Plan; per-section evidence pack supporting the annual Cybersecurity Assessment Plan self-assessment.

What CSW does not change. CSW does not perform Critical Cyber System designation, does not enforce on OT devices, does not perform deep packet inspection of OT protocols (DNP3,

IEC 61850, Modbus, OPC-UA), and does not author the Cybersecurity Architecture Design Review or the CIRP. It is the IT-side evidence engine that those governance artefacts depend on.

2. Compliance Posture Summary

The following table summarises CSW's contribution to each in-scope TSA SD requirement. Coverage is rated as:

- **Direct** — CSW is the primary technical control or evidence source for the requirement on the IT side.
- **Supporting** — CSW supplements another control with evidence or enforcement that strengthens the inspection response.
- **Out of scope** — CSW does not address the requirement; relies on other parts of the cyber programme (HR, physical, governance, OT controls).

SD Section	Topic	Coverage	CSW contribution
Section I	Applicability + definitions	Evidence Required	Regulatory determination
Section II	Critical Cyber System identification	Supporting	Inventory + scope labelling for CSW-managed workloads supporting CCS; cross-reference to your CCS register
Section II	Cybersecurity Coordinator role	Supporting	Quarterly evidence pack to the Coordinator
Section III.A	Network segmentation (IT/OT)	Direct (IT-side)	Per-site IT-side enclave segmentation; deny-by-default to IT/OT boundary firewall
Section III.A	Documented data flows	Direct (IT-side)	Observed IT → OT flow inventory + reconciliation against architecture
Section III.B	Access control measures	Direct (IT-side)	Workload-level least-privilege; quarterly access review with sunset log
Section III.C	Continuous monitoring & detection	Direct (IT-side)	Behavioural rules + flow telemetry to SIEM; conversation-graph anomalies

SD Section	Topic	Coverage	CSW contribution
Section III.D	Unpatched-system risk reduction	Direct (IT-side)	Continuous CVE inventory + compensating-control register for delayed patches
Cybersecurity Incident Response Plan	Plan, test, 24-hour CISA reporting	Direct (evidence)	Six-artefact incident reconstruction bundle
Cybersecurity Assessment Plan (CAP)	Annual implementation assessment	Direct (evidence)	Per-SD-section evidence pack with prior-period delta
Cybersecurity Architecture Design Review	Periodic architecture review	Supporting	Workload inventory + dependency graph + flow inventory as inputs to the review
Physical security	Physical access controls	Evidence Required	Facility / PACS process
Personnel & training	HR + training	Evidence Required	HR / training process
OT device-layer controls	RTUs, flow computers, PLCs, IEDs, HMIs	Evidence Required	OT-aware monitoring stack (Cisco Cyber Vision, Claroty, Nozomi, Dragos, etc.)

Posture takeaway. CSW provides direct IT-side coverage for all four substantive cybersecurity sections (III.A through III.D), the Cybersecurity Incident Response Plan evidence requirement, and the Cybersecurity Assessment Plan self-assessment. It is supporting on Section II inventory and architecture review inputs, and explicitly out of scope for OT-device-layer controls and physical/HR scope. This is consistent with CSW’s positioning as a workload-protection platform: the IT side of the SD is where it earns its place.

3. Mapping Table — TSA SD Requirement → CSW Capability → Evidence Artefact

Requirement	CSW capability	Evidence artefact (audit-ready)
Section II: Critical Cyber System identification	Workload labelling: ccs_id, ccs_function, ot_facing, site	Per-scope inventory CSV with workload identity, OS, owner, CCS reference, OT-facing flag, site, last-seen timestamp

Requirement	CSW capability	Evidence artefact (audit-ready)
Section II: Cybersecurity Coordinator	Quarterly evidence pack to the Coordinator	Standing quarterly report covering §3–§9
Section III.A: Segmentation policy	Per-site policy workspace; documented exception list; deny-by-default to boundary firewall	Workspace export per site with simulation → enforcement transition log
Section III.A: IT/OT boundary	Workload-level deny-by-default for OT-Facing-IT zone	Quarterly enforcement diff report
Section III.A: Documented data flows	Observed IT → OT flow inventory + reconciliation against architecture	IT → OT flow inventory CSV + delta report
Section III.A: New-flow change control	Continuous monitoring for new IT → OT edges	Alert log with disposition column
Section III.B: Least privilege	Workload-level identity-aware policy with rule provenance	Policy export with provenance + last-flow timestamp per rule
Section III.B: Cadence-based access review	Quarterly review of allow ruleset; sunset log for stale rules	Quarterly access review minutes + sunset log
Section III.C: Behavioural detection	Out-of-the-box + custom forensic rules tied to CCS function	Detection rule catalogue with last-modified dates and SD coverage mapping
Section III.C: Anomaly detection	Conversation-graph anomalies (new edges, port/protocol distribution change)	Anomaly alert log with investigation outcome
Section III.C: SIEM integration	Forwarding configuration to Splunk / QRadar / Sentinel / Cisco XDR	SIEM forwarding configuration + sample alert with full forensic context
Section III.C: Detection latency	Per-CCS-function MTTD report	Quarterly MTTD trend report
Section III.D: CVE inventory	Per-workload CVE list with CVSS, exploit availability, EPSS	CVE export dated to assessment period, prioritised by CCS function
Section III.D: Compensating controls	Documented in workload workspace notes; enforced via CSW policy	Compensating-control register for unpatched-but-mitigated workloads
Section III.D: Patch verification	Package inventory updates within snapshot cycle after	Diff report showing patched version + automatic CVE

Requirement	CSW capability	Evidence artefact (audit-ready)
	patch	drop
CIRP: Detection and response	Behavioural rules + quarantine policy + forensic export	Quarantine policy applied (timestamped) + six-artefact dossier
CIRP: 24-hour CISA reporting	Standard incident dossier template	Per-incident dossier ready to feed CISA submission
CAP: Annual self-assessment	Per-SD-section evidence pack with prior-period delta	Annual CAP binder organised by SD section + Senior Cybersecurity Officer sign-off summary
Cybersecurity Architecture Design Review	Workload inventory + dependency graph + observed flow inventory	Pre-review evidence pack feeding the architecture review

4. Out-of-Scope and Complementary Controls

CSW does **not** address the following SD requirements; reference the noted complementary controls.

Requirement	Owner	Complementary control
Critical Pipeline designation	TSA	Regulatory determination
Critical Cyber System designation	Operations Risk + Senior Cybersecurity Officer	Operational impact analysis
Cybersecurity Coordinator appointment	Senior Cybersecurity Officer	Organisational role assignment
Physical security at compressor and pump stations	Facilities + Physical Security	Site hardening, perimeter, CCTV
Personnel & background checks	HR + Senior Cybersecurity Officer	Background investigation, role-based training
IT/OT boundary firewall	Network/Firewall team	Boundary firewall pair (Cisco Firepower, Palo Alto, Fortinet, etc.)
OT device-layer enforcement (RTUs, flow computers, PLCs, IEDs, HMIs)	OT/Operations	OT-aware monitoring (Cyber Vision, Claroty, Nozomi, Dragos)
OT-protocol deep packet inspection (DNP3, IEC 61850, Modbus, OPC-UA)	OT/Operations	OT-aware sensors at the protocol layer

Requirement	Owner	Complementary control
Physical-to-cyber blast radius modelling	Engineering + Operations	Process safety + cyber-physical risk analysis
Pipeline operational continuity planning	Operations + Senior Cybersecurity Officer	Operational continuity plan execution

CSW's role in a complete TSA SD programme is the IT-side segmentation, monitoring, and continuous evidence engine. It explicitly defers the OT device-layer scope to the appropriate specialised stack and the governance / HR / physical scope to the appropriate owners.

5. Suggested Next Steps

For TSA-designated pipeline owner/operators considering CSW within their SD programme:

1. **Confirm CCS inputs.** Validate the Critical Cyber System list and the OT-facing IT tier definition before starting CSW scope design. CSW labels mirror this taxonomy; bad inputs produce bad evidence.
 2. **Pilot on the OT-facing IT tier first.** SCADA jump servers, historian/MES, engineering workstations, and vendor-access hosts give the highest-value wins (Section III.A IT/OT segmentation, Section III.B access control, Section III.C continuous monitoring) with the lowest deployment friction.
 3. **Pair with the existing IT/OT firewall.** CSW does not replace the IT/OT firewall. Position CSW as the IT-side hardening *up to* the firewall, with the firewall remaining the boundary control of record under Section III.A.
 4. **Wire the observed-vs-documented IT → OT flow report into the architecture review cycle.** This is the artefact most likely to surface stale documentation and undocumented operational flows.
 5. **Tie the compensating-control register to the patch programme day one.** The Section III.D requirement is most often missed when patches are delayed for operational reasons; the CSW compensating-control register is the audit-ready evidence that risk has been *reduced*, not just deferred.
 6. **Use the CAP evidence pack to drive the annual self-assessment.** The pack is structured per SD section and produces a prior- period delta automatically, which simplifies the annual review.
 7. **Plan SME review.** This document is a v1 mapping. Before being relied upon in a formal compliance engagement, route to a TSA Pipeline cybersecurity specialist for regulatory accuracy and to a current Cisco Secure Workload SE for product-capability accuracy.
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6. Disclaimer

This report and its companion reference design describe how Cisco Secure Workload capabilities can support a TSA Pipeline Security Directive cybersecurity programme on the **IT side** of the

IT/OT boundary. They are **not** legal or regulatory advice and do not constitute a finding of compliance.

Specifically:

- TSA Security Directives are issued and revised by the Transportation Security Administration; the active revision letter (02, 02A, 02B, 02C, 02D, ...) and the substantive requirements evolve. Always validate against the **current effective revision** available from TSA, against your inspector's current guidance, and against any subsequent rulemaking that supersedes the SD.
- Designation as a Critical Pipeline owner/operator is a TSA determination. Identification of Critical Cyber Systems, designation of the Cybersecurity Coordinator, the design of the Cybersecurity Incident Response Plan and the Cybersecurity Assessment Plan, Cybersecurity Architecture Design Review decisions, and incident reporting decisions remain the responsibility of the owner/ operator's Senior Cybersecurity Officer and qualified compliance counsel.
- CSW does not enforce policy on RTUs, flow computers, PLCs, IEDs, HMIs, or any other OT device, and is not certified as an OT- protocol deep-packet-inspection or architecture review tool.

This document should receive subject-matter-expert review for both TSA SD regulatory accuracy and current Cisco product capability before being used in a customer engagement.